GraphGST: Graph Generative Structure-Aware Transformer for Hyperspectral Image Classification

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***Abstract*— Transformer holds significance in deep learning (DL) research. Node embedding (NE) and positional encoding (PE) are usually two indispensable components in a Transformer. The former can excavate hidden correlations from the data, while the latter can store locational relationships between nodes. Recently, the Transformer has been applied for hyperspectral image (HSI) classification because the model can capture long-range dependencies to aggregate global features for representation learning. In an HSI, adjacent pixels tend to be homogeneous, while the NE does not identify the positional information of pixels. Therefore, PE is crucial for Transformers to understand locational relationships between pixels. However, in this area, most Transformer-based methods randomly generate PEs without considering their physical meaning, which leads to weak representations. This article proposes a new graph generative structure-aware Transformer (GraphGST) to solve the above-mentioned PE problem when implementing HSI classification. In our GraphGST, a new absolute PE (APE) is established to acquire pixels' absolute positional sequences (APSs) and is integrated into the Transformer architecture. Moreover, a generative mechanism with self-supervised learning is developed to achieve cross-view contrastive learning (CL), aiming to enhance the representation learning of the Transformer. The proposed GraphGST model can capture local-to-global correlations, and the extracted APSs can complement the spectral features of pixels to assist in NE. Several experiments with real HSIs are conducted to evaluate the effectiveness of our GraphGST. The proposed method demonstrates very competitive performance compared with other state-of-the-art (SOTA) approaches. Our source codes will be provided in the following link https://github.com/yuanchaosu/TGRS-graphGST.**

***Index Terms*— Contrastive learning (CL), graph representation learning, hyperspectral image (HSI) classification, transformer.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are high-dimensional data composed of hundreds of continuous spectral bands, collecting spatial information and spectral signatures concurrently [1], [2]. Since each pixel records a spectral signature/curve, different materials in HSIs can be distinguished by the difference between their spectral signatures [3]. HSI classification assigns a unique class label to each pixel in an HSI, and this technology has been broadly applied in many fields, such as mineral exploration, precision agriculture, vegetation monitoring, and biomedical analysis [4]. Over the past decade, researchers have commonly exploited spatial and spectral characteristics to enhance classification performance [5].

Although traditional classifiers have been used to perform HSI classification, these approaches only acquire shallow features, becoming increasingly challenging to meet users' requirements when encountering complex scenarios [6]. Deep learning (DL) has recently significantly influenced the development of HSI classification [7]. Compared with conventional classification methods, DL-based approaches can obtain better results because deep models can explore hidden characteristics in a hierarchical manner [8], [9].

Nowadays, many advanced tools and theories in the DL field have been applied to design classification approaches for HSIs, such as convolutional neural networks (CNNs) [10], residual networks (ResNets) [11], long short-term memory networks (LSTMs) [12], or generative adversarial networks (GANs) [13]. CNN-based methods can hierarchically extract features, which is beneficial to capture intrinsic correlations between pixels to assist classification [14], [15]. However, CNN-based methods often confront overfitting problems if the number of training samples is insufficient [16]. Although ResNets utilize residual connections to avoid gradient vanishing for using convolution blocks, ResNet-based approaches cannot aggregate global correlations from data effectively [17]. LSTM-based methods consider global correlations by capturing long-term dependencies in long sequences, while these

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methods easily ignore some local characteristics [18]. GANs can automatically enhance generalization by adversarial training between the generator and the discriminator [19]. However, it is difficult for GAN-based methods to find a balance between the generator and the discriminator to obtain stable results [20]. Overall, considering local and global characteristics is still a significant challenge when developing DL-based methods [21].

Graph neural networks (GNNs) are DL models that can better aggregate local correlations when handling graph-structured data [22], [23]. Recently, many researchers utilized GNNs to develop HSI classification methods, such as the spectral–spatial graph convolutional network (S^2GCN) [24], nonlocal GCN [25], multiscale dynamic GCN [26], minibatch GCN [27], spatial-spectral adaptive probability GCN [28], and graph-polarized fusion network [29]. Although these GNN-based methods adopt various strategies to realize the transformation from grid data to graph-structured data, these methods lack a global vision, leading to neglecting correlations between some pixels that are homogeneous but not spatially adjacent in an HSI [30]. Therefore, considering the development of HSI classification approaches, designing a global vision to restrain training is extremely meaningful for playing to the advantages of graph models.

Transformers are one of the most popular DL models. They are based on a structure known as the multihead self-attention (MHSA) mechanism, which acquires global dependencies better than most DL models [31]. Recently, the popularity of chat generative pre-trained Transformer (ChatGPT) is a strong testament to Transformer's feature expression capabilities, and ChatGPT can better understand contextual information through generative strategies [32]. Moreover, researchers working on DL have proposed various Transformers to solve problems in computer vision and natural language processing [33]. It is worth noting that node embedding (NE) and positional encoding (PE) are two indispensable components in a Transformer [34]. NE excavates hidden correlations from the data, while PE stores the locational relationships between nodes.

In the remote sensing community, some researchers have introduced the Transformer architecture for designing HSI classification methods, such as SpectralFormer [30], HSI Transformer network (HiT) [35], spectral–spatial Transformer network [36], local-enhanced spectral–spatial Transformer [37], spectral–spatial feature tokenization Transformer (SSFTT) [38], mask-guided spectral-wise Transformer (MST) [39], multilevel spectral–spatial Transformer [40], multibranch attention Transformer network [41], multistage fusion convolutional Transformer network [42], multiscanning-based recurrent neural network Transformer [43], global-local 3-D convolutional Transformer network [44], multiattention Transformer network [45], cascaded convolution-based Transformer network [16], masked autoencoding spectral–spatial Transformer (MAESTD) [46], spectral–spatial masked Transformer [47], spectral–spatial morphological attention Transformer (morphFormer) [48], and some lightweight Transformers [49], [50]. However, most of these Transformer-based methods combine CNNs or other deep neural networks (DNNs), drastically increasing the computational burden. Furthermore, most of them use random strategies to generate PEs without considering their physical meaning, which leads to Transformer models being unaware of the locational relationships between pixels.

A. Motivation

It is well-known that adjacent pixels in an image tend to be homogeneous [26], [27]. However, NE itself does not understand the locational relationship between pixels [34]. Hence, PE is a crucial element of Transformers for understanding the locational relationships of nodes [22]. Although many works have recently adopted Transformers for HSI classification, they normally use some random processes to generate PEs, which cannot extract the corresponding physical meaning and might lose some details of local characteristics. Moreover, most Transformer-based classification methods combine visual Transformers (ViTs) and CNNs (or other DNNs) to aggregate local correlations between pixels. As a result, the drawbacks of CNNs or other networks are inherited by these Transformer-based approaches, such as severe computational burdens caused by learning convolution kernels and over-smoothing problems caused by using patches as tokens when using ViT. Our research motivation is to solve the above-mentioned issues when adopting the Transformer architecture for HSI classification.

B. Overview

This article proposes a graph generative structure-aware Transformer (GraphGST) for HSI classification. Our GraphGST model can aggregate local-to-global correlations by integrating graph structural information into the Transformer architecture. In our newly proposed GraphGST, we first extract absolute positional sequences (APSs) at a negligible storage cost. We then consider an HSI graph where nodes represent pixels and edges capture the similarities between neighboring pixels. Next, we employ a message-passing neural network (MPNN) [51] to learn the pixels' graph structural embeddings (GSEs). Our GraphGST stacks several classic Transformer encoders, and we call each of them an encoder. The extracted GSEs serve as inputs to the first encoder, enabling all classic Transformer encoders to recognize structural interactions between pixels. Simultaneously, GrapGST adopts a generative strategy and establishes a novel Transformer encoder variant: the positional encoder. The positional encoder discerns the APSs of pixels and identifies the locational relationships between any pixel pair in a batch. These classic Transformer encoders forge fully connected (FC) graphs via the MHSA mechanism, facilitating global feature fusion and presenting pixel representations from a global perspective. In contrast, the positional encoder accentuates feature propagation between adjacent pixels, providing a local perspective on pixel representation learning. These classic encoders and the positional encoder offer different viewpoints on pixel representation learning, thus complementing each other. We employ a cross-view contrastive mechanism to facilitate the two views to supervise and complement each other

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collaboratively. Additionally, guided by the assumption that adjacent pixels tend to be homogeneous, we develop a novel positive sample selection strategy to implement self-supervised learning. Specifically, two pixels are considered positive samples in a batch if their APSs are analogous. By maximizing the mutual information of these positive samples across the two different views, our GraphGST achieves superior pixel representations. It is worth noting that conventional Transformer models will directly add PEs to NEs before inputting NEs into the MHSA mechanism. Unlike conventional Transformer models, the proposed GraphGST makes NE and PE two dependent inputs to Transformer encoders, which are more comprehensive for considering the multiple relationships between pixels. To avoid gradient vanishing, we imitate residual connections, concatenating the output of each Transformer encoder as the representative features of pixels. Finally, we combine a FC layer with a Softmax function to obtain the probability distribution of each label associated with each pixel, classifying and labeling all pixels in the HSI.

C. Contributions

The main contributions of the proposed approach can summarized as follows.

1) Graph structure learning (GSL) is integrated into an MHSA mechanism for capturing the structural interactions between pixels, thus comprehensively modeling the correlations between pixels.
2) We introduce an absolute PE (APE) method specifically designed to extract the APSs of pixels within the HSI. The APE method enables the model to identify locational correlations between any two pixels, and the APSs can be seen as the novel characteristics of pixels.
3) Contrary to ViT architectures, our GraphGST takes GSEs of pixels as tokens rather than adopting a patch-to-patch strategy to implement training. This ensures the discrimination between heterogeneous pixels.
4) The cross-view contrastive mechanism incorporates a novel positive sample selection strategy to enable local and global views to supervise and complement each other collaboratively.

II. Related Works

A. Transformers on Graphs

Let a graph be expressed as $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ represents the set of nodes, and $\mathcal{E}$ denotes the set of edges. $\mathbf{X} \in \mathbb{R}^{p\times d}$ represents node features for all nodes, and p denotes the number of nodes, and d means the dimension of node features.

The Transformer architecture is composed of an MHSA mechanism and a feed-forward neural network (FFN) [31]. From the viewpoint of GNNs, the Transformer architecture employs the MHSA mechanism to construct multiple FC graphs and capture the underlying correlations between nodes [52]. In the self-attention module, the input node features $\mathbf{X}$ are first projected to Query $\mathbf{Q}$, Key $\mathbf{K}$ and Value $\mathbf{V}$, respectively,

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_{\mathbf{Q}}, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_{\mathbf{K}}, \ \mathbf{V} = \mathbf{X}\mathbf{W}_{\mathbf{V}} \tag{1}$$

where $\mathbf{W}_{\mathbf{Q}}$, $\mathbf{W}_{\mathbf{K}}$, and $\mathbf{W}_{\mathbf{V}}$ are trainable parameters, respectively. The self-attention can be calculated by the following formula:

$$\text{Attention}(\mathbf{X}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_{\mathbf{Q}}}}\right)\mathbf{V} \tag{2}$$

where d_Q represents the dimension of $\mathbf{Q}$. It is common to use multihead attention, which concatenates multiple instances of (2) to improve self-attention learning. Next, the output of the MHSA mechanism is followed by a skip-connection and an FFN, which jointly compose a Transformer encoder, as illustrated below

$$\begin{aligned}\mathbf{X}' &= \mathbf{X} + \text{Attention}(\mathbf{X})\\ \mathbf{X}'' &= \text{FFN}\left(\mathbf{X}'\right).\end{aligned} \tag{3}$$

Multiple Transformer encoders are typically stacked as a deep model to learn node representations, regardless of their surrounding structures in the graph [22]. It is thus necessary to incorporate the graph structural information into the Transformer. Following the view of GSL in [53], the attention in (2) can be re-expressed as:

$$\text{Attention}(\mu) = \sum_{\nu\in\mathcal{V}} \frac{\kappa_{\text{graph}}(S_G(\mu), S_G(\nu))}{\sum_{\omega\in\mathcal{V}} \kappa_{\text{graph}}(S_G(\mu), S_G(\omega))}\mathcal{F}(\mathbf{x}_\nu) \tag{4}$$

where $S_G(\mu)$ represents a subgraph of G and centered at node μ, $\mu, \nu \in \mathcal{V}$. $\kappa_{\text{graph}}(\cdot,\cdot)$ is introduced to measure the correlation between two subgraphs. $\mathbf{x}_\nu$ denotes the features of node ν, and $\mathcal{F}(\mathbf{x}_\nu) = \mathbf{W}_{\mathbf{V}}\mathbf{x}_\nu$ denotes the linear value function. This enhanced self-attention mechanism considers both attribute similarity and structural similarity between nodes. It thus produces more expressive node representations compared to the conventional self-attention mechanism. $\kappa_{\text{graph}}(\cdot,\cdot)$ can be any kernel that compares a pair of subgraphs. For instance, the form of κ_{graph} can be represented as follows:

$$\kappa_{\text{graph}}\left(S_G(\mu), S_G(\nu)\right) = \exp\left(\langle\varphi(\mu, G), \varphi(\nu, G)\rangle/\sqrt{d_{\text{out}}}\right) \tag{5}$$

where $\varphi(\mu, G)$ is a structure extractor that extracts vector representations of some subgraph centered at node μ. $\langle\cdot,\cdot\rangle$ represents the inner product of two structural vector representations. $\exp(\cdot)$ denotes the element-wise exponentiation. d_{out} represents the dimension of structural vector representations.

B. Contrastive Learning

Contrastive learning (CL) is a branch of self-supervised learning that is adept at deriving supervised signals from the data to improve the generalization ability of models [54]. In recent years, CL has been developed as one of the most popular self-supervised learning strategies due to flexible expandability [55]. Nowadays, CL has been successfully used in many fields, such as visual representation [56], person re-identification [57], and action recognition [58].

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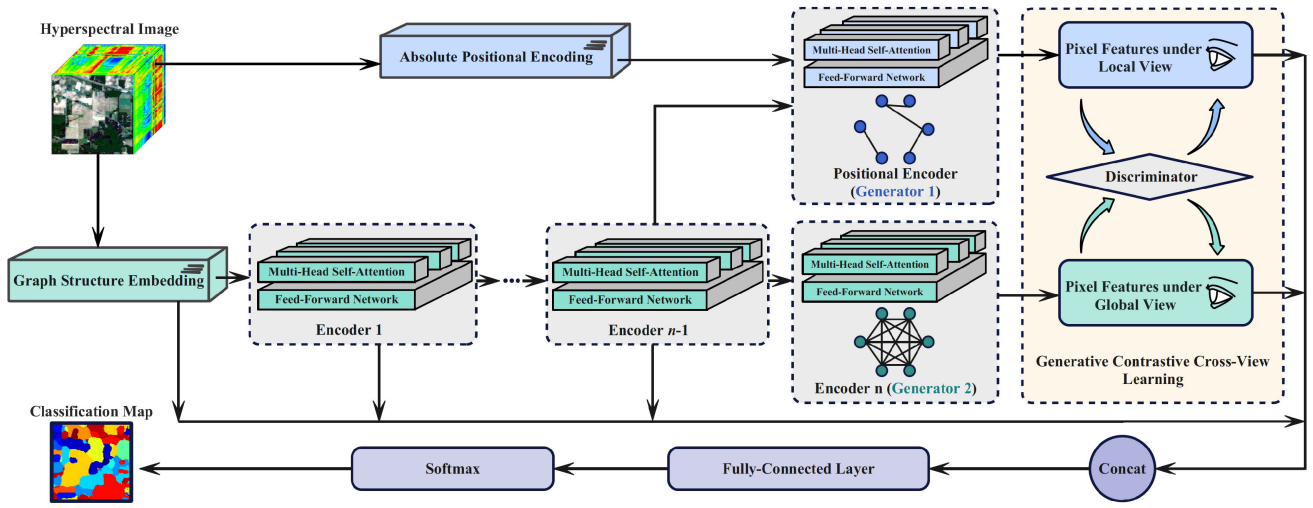


Fig. 1. Flowchart of the proposed GraphGST. The APE can acquire the APSs for each pixel. An MPNN is employed to implement the GSE of pixels. GraphGST contains n classic Transformer encoders and a positional encoder. Moreover, the generative strategy refers to two generators and a discriminator for serving the GCCVL. Finally, an FC layer combines with a Softmax function to label pixels.

In normal circumstances, CL first extracts positive and negative samples from data and then maximizes the similarities between positive samples while minimizing the similarities between negative samples [59]. Furthermore, users can obtain discriminative representations (even without abundant labels) when using CL [60]. The InfoNCE is a common loss function in CL, considering its superior capacity in modeling pairwise correlations [57]. The loss function of InfoNCE can be formulated as

$$\mathcal{L}_{\text{Info}} = -\log \frac{\mathcal{S}\left(\mathbf{x}_{\mu}^{i}, \mathbf{x}_{\mu}^{j}\right)}{\sum_{i \neq j}\left(\mathcal{S}(\mathbf{x}_{\mu}^{i}, \mathbf{x}_{v}^{i}) + \mathcal{S}(\mathbf{x}_{\mu}^{i}, \mathbf{x}_{v}^{j})\right)} \tag{6}$$

where i and j are utilized to distinguish two viewpoints. u and v represent different samples, and $\mathcal{S}(\cdot, \cdot)$ represents the similarity measurement between a pair of samples. Equation (6) is a multiview pattern, and viewpoint information is adopted as the indicator to group positive samples.

After calculating the InfoNCE loss, models can usually be updated via gradients [58]. Empirically, the learned multiview features are more informative if more viewpoints are included during the training of deep models [61]. Moreover, CL can also be extensible to other modalities to improve the sample representation learned by setting contrastive losses [60].

III. Proposed Approach

The proposed GraphGST is an inductive learning model that establishes three mechanisms: APE, graph-structure-aware attention (GSAA), and generative contrastive cross-view learning (GCCVL). The APE is developed to obtain the absolute positional information for each pixel in an HSI, enabling the model to capture the locational relationships between any pair of pixels. The GSAA is designed for recognizing structural interactions among pixels. The GCCVL enables the local and global views to supervise and complement each other collaboratively, achieving self-supervision to enhance the representation. Finally, we employ an FC layer and a Softmax function to classify pixels in an HSI. The framework of our GraphGST is illustrated in Fig. 1.

A. Absolute PE

We design the APE to obtain the APSs of each pixel, used as the input of the positional encoder. An example is given in Fig. 2 to illustrate the procedure of obtaining the APSs of a pixel. Assuming that s and r define the size of an HSI, s denotes the height of the image, and r defines its width, we have $p = s \times r$. We first use a patch-wise scheme to segment the HSI to relieve the computational burden. Specifically, assuming that the size of a patch is $a \times a$. Let & be the number of patches, the corresponding calculation method is as follows:

$$\& = \text{Ceil}\left(\frac{s}{a}\right) \times \text{Ceil}\left(\frac{r}{a}\right) \quad \forall\ \& \ll s \times r \tag{7}$$

where $\text{Ceil}(\cdot)$ is the nearest integer function that rounds a number in the direction of positive infinity.

Afterward, to distinguish these patches, we label them in order. As seen in Fig. 2, these patches are assigned as $1, 2, \ldots, \&$, respectively. Subsequently, to obtain the absolute positional information of each pixel, we set a sliding window and use $b \times b$ to define the window size. The sliding window places the target pixel in the central position. As observed in Fig. 2, within the sliding window, the patch labels are 1, 1, 2, 2, 2, 2, 2, 2, 3, 3, 11, 11, 11, 12, 12, 12, 12, 12, 12, 12, 12, 12, 13, 13, and 13. The unique patch labels are 1, 2, 3, 11, 12, and 13, with corresponding frequencies of 2, 6, 2, 3, 9, and 3, respectively. We utilize these frequencies as patch-label weights for a pixel. Considering all patches, we can obtain a sparse APS $[2, 6, 2, 0, 0, 0, 0, 0, 0, 0, 3, 9, 3, 0, \ldots, 0] \in \mathbb{R}^{1 \times \&}$ for the pixel. Zeros mean that the corresponding patches do not appear in the sliding window. A positional sequence is also illustrated at the bottom of Fig. 2. The purpose of APE is to provide weights of patch labels for each pixel and form APSs, which will be an input feature of the positional encoder.

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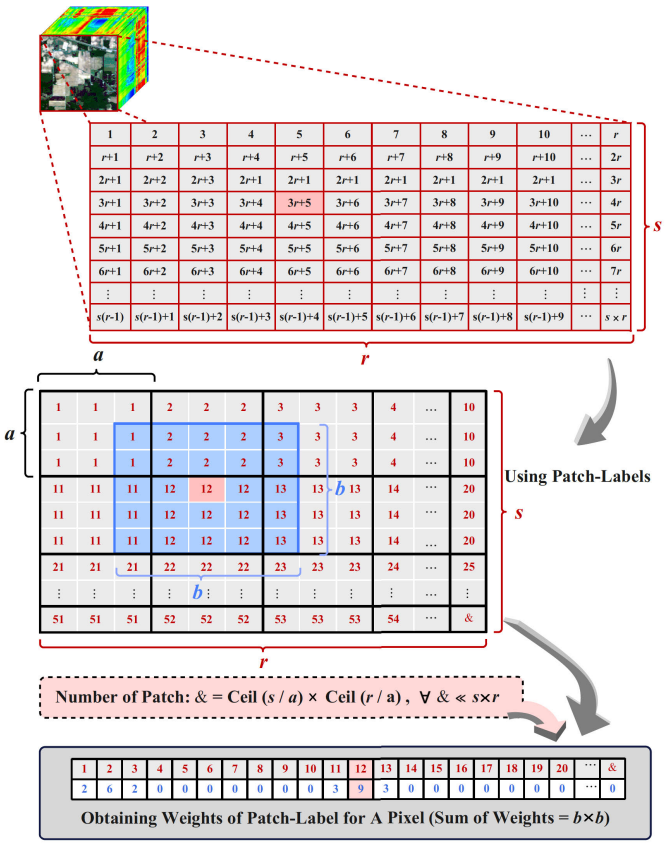


Fig. 2. Illustration of the process for obtaining patch-label weights for a pixel. We use r and s to define the HSI's width and height. In this example, we pay attention to the $(3r + 5)$th pixel and obtain its patch-label weights using APE. The patch-label weights are taken as the APS and will be an input feature of the subsequent positional encoder.

The proposed APE can obtain low-dimensional APSs due to $\& \ll p$. Therefore, for large-scale HSIs, users do not encounter severe computational burdens when using the proposed APE method. Although the APE reduces the dimension of the APSs, each nonzero element in the sequence can reflect the weight of the pixel on the corresponding patch. Hence, the obtained positional sequences can accurately express the absolute positional information of pixels in an HSI, which is practical and significant for applying GraphGST to process large-scale HSIs.

B. Graph-Structure-Aware Attention

In this article, we regard an HSI as a graph structure dataset where the nodes correspond to the pixels, and there are initial edges between adjacent nodes. Utilizing the constraint scheme in [62], we also set a window to limit the neighborhood range. The target pixel is placed at the center of the window, and other pixels in the window are its neighboring nodes. The window size equals the sliding window size $b \times b$ in Section III-A. Herein, we employ the MPNN to learn the target node's GSEs by aggregating its neighbor's features within the pixel graph. The procedure can be written as

$$\mathbf{h}_v = \sigma\left(\sum_{\mu \in \mathcal{N}_v} \mathbf{e}_{\mu v} \mathbf{x}_\mu \mathbf{W}^0\right) \tag{8}$$

where $\mathbf{h}_v \in \mathbb{R}^{1\times d}$ denote the GSEs of pixel v, $\mathcal{N}_v$ represents the set of pixels in the window and can be seen as the set of pixel v and its neighboring pixels. $\mathbf{e}_{\mu v}$ denotes the weight between nodes μ and v. $\sigma(\cdot) = \max(0, \cdot)$ represents the rectified linear unit (ReLU) activation function. $\mathbf{X} = \{\mathbf{x}_\mu\}_{\mu=1}^{p} \in \mathbb{R}^{p\times d}$ represents the normalized spectral feature matrix of pixels. In (8), $\mathbf{W}^0 \in \mathbb{R}^{d\times d}$ represents a trainable weight matrix. After using (8), each pixel will acquire the corresponding GSEs. During the implementation of GraphGST, we adopt a batch-wise strategy to train the model. We assume that the batch size (bs) is c. In a batch, GSEs of c pixels can constitute a matrix $\mathbf{H}^{(0)} \in \mathbb{R}^{c\times d}$.

The proposed GraphGST incorporates n classic Transformer encoders, as shown in Fig. 1. As observed, the first classic transformer encoder (Encoder 1) takes in the pixels' GSEs $\mathbf{H}^{(0)}$ and outputs $\mathbf{H}^{(1)} \in \mathbb{R}^{c\times d}$. Since $\mathbf{H}^{(0)}$ can express graph structural information of pixels, Encoder 1 can capture the structural interaction between any pair of pixels in a batch. Next, to better accomplish the downstream tasks, we stack multiple transformer encoders (Encoder 2, Encoder 3 , . . . , Encoder n) to learn more abstract features: $\mathbf{H}^{(2)}, \mathbf{H}^{(3)}, \ldots, \mathbf{H}^{(n)}$. Following (1), (2), and (4), the representation learning within Encoder $m+1$ $(m = 0, 1, \ldots, n-1)$ can be denoted as:

$$\begin{aligned} \widehat{\mathbf{Q}}^{(m)} &= \mathbf{H}^{(m)}\mathbf{W}_Q^{(m)} \\ \widehat{\mathbf{K}}^{(m)} &= \mathbf{H}^{(m)}\mathbf{W}_K^{(m)} \\ \widehat{\mathbf{V}}^{(m)} &= \mathbf{H}^{(m)}\mathbf{W}_V^{(m)} \end{aligned} \tag{9}$$

where $\widehat{\mathbf{Q}}^{(m)}$, $\widehat{\mathbf{K}}^{(m)}$ and $\widehat{\mathbf{V}}^{(m)}$ represent Query, Key, and Value, respectively. $\mathbf{W}_Q^{(m)}$, $\mathbf{W}_K^{(m)}$, and $\mathbf{W}_V^{(m)}$ are trainable weights. Next, the self-attention is rewritten as

$$\text{Attention}(\mathbf{H}^{(m)}) = \text{Softmax}\left(\frac{\widehat{\mathbf{Q}}^{(m)}\widehat{\mathbf{K}}^{(m)\top}}{\sqrt{d_Q}}\right)\widehat{\mathbf{V}}^{(m)} \in \mathbb{R}^{c\times d_Q}. \tag{10}$$

Multihead attention concatenates multiple instances of (10). Next, an FFN is employed to update the output of the MHSA mechanism. The following formulas describe the update procedure:

$$\begin{aligned} \mathbf{H}'^{(m)} &= \text{LayerNorm}\left(\mathbf{H}^{(m)} + \text{Attention}\left(\mathbf{H}^{(m)}\right)\mathbf{W}'^{(m)}\right) \\ \mathbf{H}^{(m+1)} &= \text{LayerNorm}\left(\mathbf{H}'^{(m)} + \mathbf{H}'^{(m)}\mathbf{W}''^{(m)}\right) \end{aligned} \tag{11}$$

where $\mathbf{H}^{(m+1)}$ is the output of Encoder $m+1$. $\mathbf{W}'^{(m)}$ and $\mathbf{W}''^{(m)}$ are learnable weights, and LayerNorm $(\cdot)$ can accelerate convergence. Fig. 3(a) provides a diagrammatic drawing of a classic Transformer encoder. For the sake of description, we uniformly refer to $\mathbf{H}^{(0)}, \mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(n-1)}$ as latent features of pixels.

In addition to n classic Transformer encoders, the proposed GraphGST contains a Transformer variant called a positional encoder, which can capture locational relationships between pixels. Specifically, in the positional encoder, the Query and Value are determined by the $\widetilde{\mathbf{H}}$ and $\mathbf{H}^{(n-1)}$, respectively,

$$\widetilde{\mathbf{Q}} = \widetilde{\mathbf{H}}\mathbf{W}_{\widetilde{Q}}, \quad \widetilde{\mathbf{V}} = \mathbf{H}^{(n-1)}\mathbf{W}_{\widetilde{V}} \tag{12}$$

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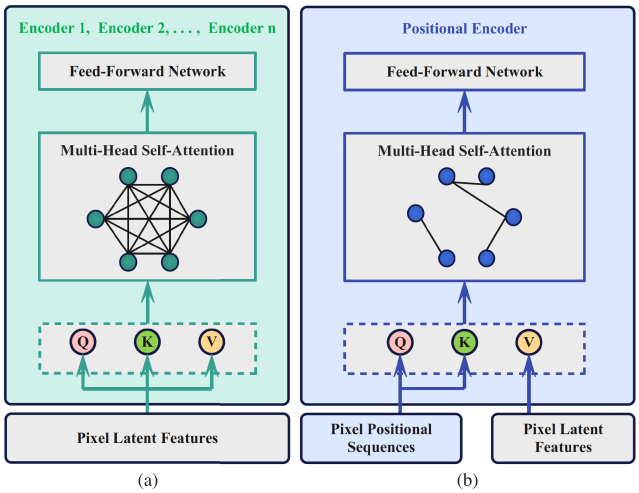


Fig. 3. Diagrammatic drawing of transformer encoders in GraphGST. (a) Encoders 1 to n. (b) Positional encoder.

where $\widetilde{\mathbf{H}} \in \mathbb{R}^{c\times\&}$ is the APSs of pixels, where & is the dimension of sequences. $\mathbf{W}_{\widetilde{Q}}$, and $\mathbf{W}_{\widetilde{V}}$ are learnable weights. Afterward, the self-attention in the positional encoder can be written as

$$\mathbf{S} = \text{Softmax}\left(\frac{\widetilde{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\top}}{\sqrt{d_{\mathrm{Q}}}}\right)$$
$$\text{Attention}\left(\widetilde{\mathbf{H}}, \mathbf{H}^{(n-1)}\right) = \sigma\left(\mathbf{S}\widetilde{\mathbf{V}}\right) \in \mathbb{R}^{c\times d_Q}. \tag{13}$$

The attention operator of the positional encoder can be seen as a variant of the original attention operator in (10), specifically tailored to capture locational relationships among pixels. Next, we adopt an FFN to update the output of MHSA in the positional encoder

$$\widetilde{\mathbf{H}}' = \text{LayerNorm}\left(\mathbf{H}^{(n-1)} + \text{Attention}(\widetilde{\mathbf{H}}, \mathbf{H}^{(n-1)})\widetilde{\mathbf{W}}'\right)$$
$$\overline{\mathbf{H}} = \text{LayerNorm}\left(\widetilde{\mathbf{H}}' + \widetilde{\mathbf{H}}'\widetilde{\mathbf{W}}''\right). \tag{14}$$

Fig. 3(b) shows the architecture of the positional encoder, and we can see that the positional Encoder has two inputs. Comparing Fig. 3(a) and (b), we can observe that the GSEs mainly determine the relationships between pixels in Encoders 1 to n, while the absolute positional information determines the relationship between pixels in the positional encoder. Since the pixels of a batch are randomly selected, the graphs built by the MHSA of the positional encoder are much sparser than the graphs from Encoders 1 to n.

C. Generative Contrastive Cross-View Learning

Encoders 1 to n concentrate on pixel GSEs and can establish FC graphs. Conversely, the positional encoder utilizes APSs of pixels to establish non-FC graphs, where nodes still denote pixels, and edges exist only between adjacent pixels. Therefore, Encoders 1 to n aggregate features from a global view, while the positional encoder performs feature fusion from a local view. We incorporate a generative strategy and self-supervised CL to form the GCCVL, enabling local and global views to complement and supervise each other to learn more discriminative pixel representations.

Typically, given a pixel under the global view, we can simply define its features learned by the local view as the positive sample. For example, in (6), the two views of a sample serve as positive pairs. However, considering that adjacent pixels are highly correlated, we propose a new positive selection strategy where two nodes are positive samples of each other if they are spatially adjacent. Specifically, we employ cosine similarity and threshold setting to acquire positive samples. The cosine similarity is expressed as

$$\text{Similarity}(\widetilde{\mathbf{h}}_\mu, \widetilde{\mathbf{h}}_\nu) = \frac{\widetilde{\mathbf{h}}_\mu \cdot \widetilde{\mathbf{h}}_\nu^{\top}}{\|\widetilde{\mathbf{h}}_\mu\|\|\widetilde{\mathbf{h}}_\nu\|} \tag{15}$$

where $\widetilde{\mathbf{h}}_\mu$ and $\widetilde{\mathbf{h}}_\nu \in \mathbb{R}^{1\times\&}$ represent APSs of pixels μ and ν. Herein, pixels μ and ν can be any pair of pixels in a batch. $\gamma \in [0, 1]$ denotes a hyper-parameter that can be obtained by grid-searching. If the cosine similarity between nodes exceeds the threshold γ, the two nodes constitute a positive sample pair.

Let $\mathcal{Z}$ be the positive sample set of a batch and used to model a CL task. Naturally, we present the following loss function:

$$\ell_{\text{GCCVL}} = -\frac{1}{2|\mathcal{Z}|}\left(\sum_{(\mu,\nu)\in\mathcal{Z}} \mathbb{E}_{(\mathbf{H}^{(n)},\overline{\mathbf{H}})}\left[\log \Psi\left(\mathbf{h}_\mu^{(n)}, \overline{\mathbf{h}}_\nu\right)\right] + \sum_{(\mu,\nu)\in\mathcal{Z}} \mathbb{E}_{(\mathbf{H}^{(n)},\overline{\mathbf{H}})}\left[\log \Psi\left(\mathbf{h}_\nu^{(n)}, \overline{\mathbf{h}}_\mu\right)\right]\right) \tag{16}$$

where the objective of the CL-based loss function is to maximize the mutual information between local and global views of each positive sample pair. Herein, $\Psi(\cdot,\cdot)$ represents the discriminator. $\mathbf{h}_\mu^{(n)}$ (or $\mathbf{h}_\nu^{(n)}$) is the μth (or νth) row vector of $\mathbf{H}^{(n)} \in \mathbb{R}^{c\times d}$, and it means the feature representation of pixel μ (or ν) under global view. Likewise, $\overline{\mathbf{h}}_\mu$ (or $\overline{\mathbf{h}}_\nu$) is the μth (or νth) row vector of $\overline{\mathbf{H}} \in \mathbb{R}^{c\times d}$, and it denotes the feature representation of pixel μ (or ν) under the local view. The discriminator $\Psi(\cdot,\cdot)$ can be formulated as

$$\Psi\left(\mathbf{h}_\mu^{(n)}, \overline{\mathbf{h}}_\nu\right) = \sigma\left(\mathbf{h}_\mu^{(n)}\mathbf{W}\overline{\mathbf{h}}_\nu^{\top}\right) \tag{17}$$

where $\mathbf{W} \in \mathbb{R}^{d\times d}$ is the weight matrix, and the activation function $\sigma(\cdot)$ is the sigmoid function. Using (17), we can get the probability that $\mathbf{h}_\mu^{(n)}$ and $\overline{\mathbf{h}}_\nu$ are positive samples of each other. Minimizing ℓ_{GCCVL} enables local and global views to be complementary and able to supervise each other, thus facilitating the maximization of mutual information from adjacent pixels.

Equation (16) can be seen as an improved version of (6). Specifically, the similarity measurement $\mathcal{S}(\cdot,\cdot)$ of (6) is transformed as the discriminator $\Psi(\cdot,\cdot)$. This transformation changes maximizing similarities of positive pairs to maximizing mutual information of positive pairs. In addition, in (6), two different views of one sample are typically seen as positive pairs. However, in (16), considering adjacent pixels tend to be homogeneous in HSIs, the adjacent pixels are seen as positive pairs.

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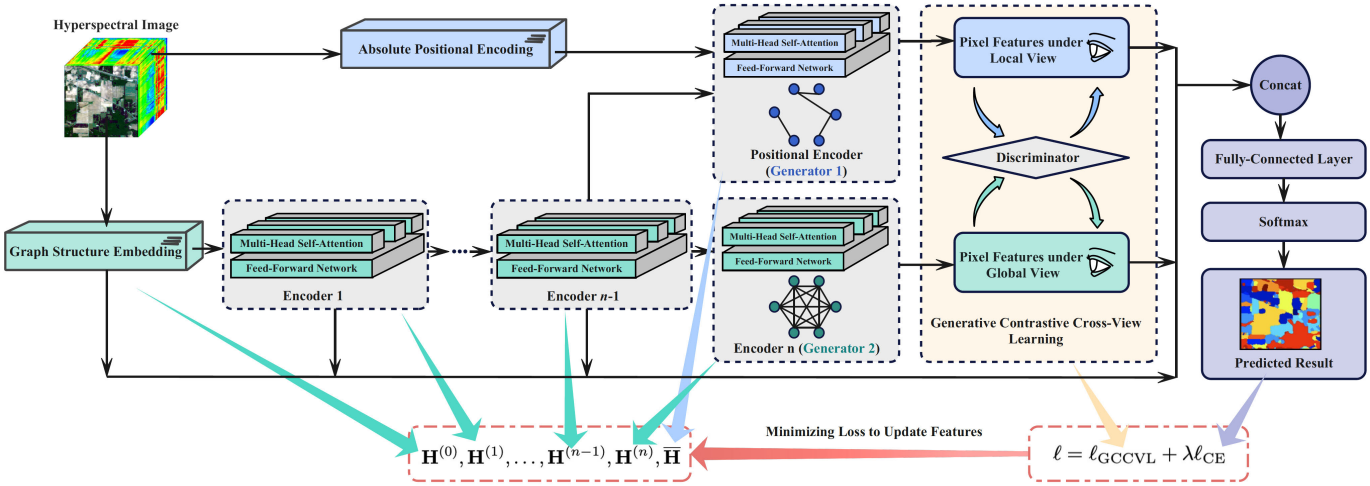


Fig. 4. Illustration of updating features and optimization in the proposed GraphGST. The GCCVL can enable local and global views to complement and collaboratively supervise each other, which achieves the short-long range dependencies for enhancing feature representation.

D. Loss Function

After implementing GSL, Encoder 1, Encoder 2, ..., Encoder n, and the positional encoder, we obtain $\mathbf{H}^{(0)}$, $\mathbf{H}^{(1)}$, $\mathbf{H}^{(2)}, \ldots, \mathbf{H}^{(n)}$, and $\overline{\mathbf{H}}$, respectively. To avoid the vanishing gradient problem, we imitate the residual connection and concatenate these outputs as high-level feature representations of pixels

$$\overrightarrow{\mathbf{H}} = \|(\mathbf{H}^{(0)}, \mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(n)}, \overline{\mathbf{H}}) \tag{18}$$

where $\overrightarrow{\mathbf{H}}$ denotes high-level features, and $\|((\cdot))$ represents the concatenate operation. Then, we use an FC layer and a Softmax function to calculate the probability distributions of pixels as follows:

$$\mathbf{Y} = \mathrm{Softmax}\left(\overrightarrow{\mathbf{H}}\mathbf{W}'''\right) \tag{19}$$

where $\mathbf{Y}$ denotes the predicted results, and $\mathbf{W}''' \in \mathbb{R}^{(n+2)d \times \xi}$ represents the weight matrix. ξ represents the number of pixel classes. Additionally, we adopt the cross-entropy (CE) loss function to calculate the classification loss

$$\ell_{\mathrm{CE}} = -\sum_{\mu \in \Lambda} \sum_{\beta=1}^{\xi} \hat{y}_{\mu\beta} \log(y_{\mu\beta}) \tag{20}$$

where Λ denotes the set of node indices. $\hat{y}_{\mu\beta}$ is an indicator variable, and $y_{\mu\beta}$ represents the predicted probability score of the μth node under the βth class. For the proposed GraphGST, the overall loss function can be written as

$$\ell = \lambda \ell_{\mathrm{CE}} + \ell_{\mathrm{GCCVL}} \tag{21}$$

where λ is the balancing hyper-parameter. Fig. 4 visualizes the updating features and the optimization.

A pseudocode of the proposed GraphGST is provided in Algorithm 1. In Line 1, $\widetilde{\mathbf{H}}$ is obtained by the APE described in Section III-A. Lines 2−20 carry out the training procedure of GraphGST, corresponding to Sections III-B and III-C. Finally, we can use the trained GraphGST model to predict the labels of test samples.

IV. Experimental Results and Analysis

A. Data Description

In this article, we use four real HSIs to assess the effectiveness and competitiveness of our GraphGST, including the Indian Pines,[1] WHU-Hi-LongKou,[2] Houston University 2013,[3] and Houston University 2018[4] datasets. The four real HSIs are common public datasets, and they include their ground-truths (GTs) for evaluating classification performance.

1) The Indian Pines dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), showcasing a rural region in Northwest Indiana, USA. The dataset comprises 145 × 145 pixels and encompasses 200 spectral bands that span from 0.2 to 2.4 μm. Additionally, the Indian Pines image contains reference data, including 16 distinct classes, which are instrumental for assessing classification results.
2) The WHU-Hi-LongKou dataset was sourced using an unmanned aerial vehicle (UAV), capturing imagery of an agricultural zone in Hubei province, China. The HSI within the dataset contains 270 spectral bands from 0.4 to 1.0 μm, and the data size is 550 × 400 pixels [63]. The scenario covers nine land-over classes, and the GT map is published for testing classification performance.
3) The Houston University 2013 dataset was gathered using a compact airborne spectrographic imager (CASI-1500 sensor), and the scenario covers the campus of the University of Houston, located in Houston, TX, USA. In 2013, the National Center for Airborne Laser Mapping (NCALM) provided the data for the data fusion contest of the IEEE Geoscience and Remote Sensing Society (GRSS). The Houston University 2013 dataset is highly challenging, and the training and test samples

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes
[2] http://rsidea.whu.edu.cn/e-resource_WHUHi_sharing.htm
[3] https://hyperspectral.ee.uh.edu/?page_id=459
[4] https://hyperspectral.ee.uh.edu/?page_id=1075

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Algorithm 1 Pseudocode of the Proposed GraphGST

Input: Pixels' spectral features: $\mathbf{X}$; Training Samples' Labels: $\hat{\mathbf{Y}}$; Threshold: α
Output: Labels of test samples.

$\widetilde{\mathbf{H}} \leftarrow$ Obtain pixels' APSs by the designed APE strategy.
//∗ *Training Procedure* ∗//
for *Epochs* **do**
  for *Batches* **do**
    $\mathbf{H}^{(0)} \leftarrow$ Use $\mathbf{X}$ to learn pixels' GSEs via (8).
    Let $m = 1$
    repeat
      $\mathbf{H}^{(m)} \leftarrow$ Use $\mathbf{H}^{(m-1)}$ and Encoder m to learn pixel features under global view via (9)-(11).
      $m = m + 1$
    until $m = n$;
    $\overline{\mathbf{H}} \leftarrow$ Use $\mathbf{H}^{(n-1)}$ and $\widetilde{\mathbf{H}}$ to learn pixel features under local view via (12)-(14).
    $\mathcal{Z} \leftarrow$ Use $\widetilde{\mathbf{H}}$ to construct a positive sample set via (15) and threshold γ.
    $\ell_{\mathrm{GCCVL}} \leftarrow$ Use $\overline{\mathbf{H}}$, $\mathbf{H}^{(n)}$ and $\mathcal{Z}$ to calculate the CL loss via (16).
    $\overrightarrow{\mathbf{H}} \leftarrow$ Concatenate $\mathbf{H}^{(0)}, \mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(n)}, \overline{\mathbf{H}}$ to obtain high-level pixel representations via (18).
    $\mathbf{Y} \leftarrow$ Use $\overrightarrow{\mathbf{H}}$ to predict classification results via (19).
    $\ell_{\mathrm{CE}} \leftarrow$ Use $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ to calculate classification loss via (20).
    $\ell \leftarrow$ Obtain total loss via (21).
    Minimize ℓ and update $\mathbf{H}^{(0)}, \mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(n)}$ and $\overline{\mathbf{H}}$.
  end
end
//∗ *Testing Procedure* ∗//
Predict labels of the test samples using the trained GraphGST model.

are fixed by NCALM. The image comprises 15 land-cover classes and is composed of 349 × 1905 pixels, encompassing 144 spectral bands that range from 0.38 to 1.05 μm.

4) The Houston University 2018 dataset was also facilitated by the University of Houston, capturing an urban expanse exceeding 4 km^2. The dataset was also obtained by the CASI-1500 sensor and provided by the NCALM for the data fusion contest of the GRSS in 2018. The data size is 601 × 2384 pixels, and 20 land-cover categories are surveyed and provided in the GT. The dataset has 48 channels, while its spectral range is the same as the Houston University 2013 data.

B. Experimental Setup

1) Evaluation Metrics: Four metrics are used for evaluating the classification performances of the proposed GraphGST, including class accuracy (CA), overall accuracy (OA), average accuracy (AA), and κ coefficient. The four metrics are often used for quantitative analysis in comparison experiments.

2) Baseline: We compare the proposed GraphGST with seven SOTA baselines.

1) FuNet [27] allows the training of large-scale GCNs in a mini-batch fashion. It extracts the HSI features by locally preserving the graph structure in one batch. FuNet has three variants: FuNet-A, FuNet-M, and FuNet-C. We chose the best-performing one, FuNet-C, for comparison.
2) NSCKL [64] is a shallow model that proposes a new kernel-based iterative filter to capture local dependencies and develops a fast spectral clustering to capture global correlations.
3) HiT [35] is a ViT-based model that develops a spectral-adaptive 3-D convolution projection module and convolution permutator to retrieve subtle spatial-spectral discrepancies.
4) SpectralFormer [30] is a novel and generic ViT-based baseline network focusing on spectrometric characteristics. SpectralFormer has two variants, pixel-wise and patch-wise. We choose the patch-wise, which performs better generally.
5) MAEST [46] is a ViT-based encoder-decoder model that uses a masking auto-encoding strategy to dynamically uncover the most robust features and employs Transformer decoders to reconstruct these features.
6) morphFormer [48] is a spectral–spatial model that uses spectral and spatial morphological convolution operations (in conjunction with the attention mechanism) to improve the interaction between the structural and shape information of the HSI token and the CLS token.
7) SSFTT [38] is a CNN-based method that combines a 3-D convolution layer and a 2-D convolution layer to extract low-level features and employs a Gaussian weighted feature tokenizer and the Transformer encoder for representation learning.

For these baseline models, we tune the parameters of each method based on recommended settings in the published codes to ensure their best performance.

3) Implementation Details: The proposed GraphGST is implemented with the DL library PyTorch. The Python and PyTorch versions are 3.8.10 and 1.9.0, respectively. We use the ReLU activation function and Adam optimizer [65] by default. All experiments are conducted on a Windows server with a graphics processing unit (GPU) (NVIDIA GeForce RTX 4090).

C. Hyper-Parameter Analysis

The proposed GraphGST involves some hyper-parameters that influence the classification performance, including bs, learning rate (lr), dropout rate (dr), training epochs (te), the size of patches (a), the size of sliding windows (b), the number of classic Transformer encoders (n), the threshold value for selecting positive samples (γ), and the weight of CE loss (λ). To thoroughly investigate the impact of each hyper-parameter

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TABLE I

HYPER-PARAMETERS PROVIDING BEST ACCURACY FOR THE PROPOSED GRAPHGST ON ALL DATASETS

Dataset	bs	lr	dr	te	a	b	n	γ	λ
Indian Pines	64	5e-4	0.2	400	5	11	9	0.1	5
WHU-Hi-LongKou	512	5e-4	0.5	400	7	11	3	0.2	10
Houston2013	512	5e-4	0.4	400	5	9	3	0.2	5
Houston2018	64	5e-4	0.5	800	7	9	9	0.1	10

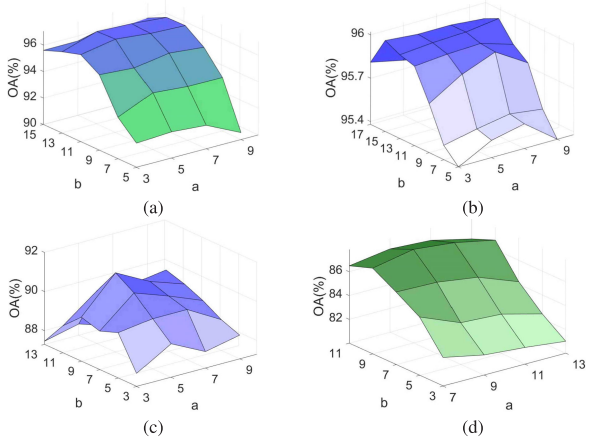


Fig. 5. OA (%) of the proposed GraphGST versus the patch size a and sliding window size b on different datasets. (a) Indian Pines dataset. (b) WHU-Hi–LongKou dataset. (c) Houston University 2013 dataset. (d) Houston University 2018 dataset.

on the classification results, a grid search strategy is employed where one hyper-parameter is varied while keeping the other hyper-parameters fixed. Table I summarizes the values of all hyper-parameters with the best accuracy on four datasets. Among these hyper-parameters, a, b, n, and γ are specific to the proposed GraphGST. Therefore, in the subsequent experiments, we will detail the impact of these four hyper-parameters on the experimental results of GraphGST and analyze the reasons.

1) Impacts of Patch Size a and Sliding Window Size b: a and b are key parameters for obtaining effective APSs and GSEs for each pixel. a determines the granularity in assigning labels to patches and the dimension of APSs. b determines the density level of the APSs and the maximum distance from neighboring pixels to the target pixel in the pixel graph. When b is small, the target pixel may not be able to aggregate enough graph structural information. Conversely, when b is too large, it can lead to over-smoothing issues. Therefore, an appropriate value of b is crucial for achieving a satisfactory classification result. Fig. 5 presents the OA (%) comparisons for various values of a and b on four datasets. Fig. 5(a) displays the comparisons on the Indian Pines dataset. In this dataset, we set a in the range $[3, 5, \ldots, 9]$ and b in the range $[5, 7, \ldots, 15]$. It can be observed that the OAs are not significantly affected by variations in a. This observation verifies the APE's capability to acquire low-dimensional APSs. In contrast, we observe that b significantly impacts OAs. Specifically, OAs improve when b increases from 5 to 11, and OAs decrease when b grows from 11 to 15. This observation confirms the importance of graph structural information to the model. Finally, we set $a = 5$ and $b = 11$ for the Indian Pines dataset to ensure the best performance. Fig. 5(b)–(d) display the comparisons on the WHU-Hi-LongKou, Houston University 2013, and Houston University 2018 datasets, respectively. From Fig. 5(b)to (d), we observe situations similar to those in Fig. 5(a). These experimental results reconfirmed our hypothesis. Moreover, we record the values of a and b on four different datasets in Table I.

TABLE II

OA (%), AA (%), AND κ (%) OF THE PROPOSED GRAPHGST VERSUS THE THRESHOLD VALUE γ ON ALL DATASETS

Dataset	γ	0	0.1	0.2	0.3	0.4	0.5
Indian Pines	OA	96.82	**97.04**	96.71	96.21	96.30	96.25
	AA	98.10	**98.17**	98.17	98.08	98.07	98.01
	κ	96.35	**96.49**	96.23	95.66	95.84	95.74
WHU-Hi-LongKou	OA	96.06	96.10	**96.13**	96.09	96.02	96.00
	AA	96.25	96.28	**96.33**	96.43	96.34	96.18
	κ	94.87	94.92	**94.96**	94.91	94.95	94.90
Houston2013	OA	90.34	90.60	**90.65**	90.37	90.38	90.25
	AA	91.68	91.93	**92.00**	91.56	91.61	91.46
	κ	89.52	90.02	**90.11**	89.55	89.56	89.64
Houston2018	OA	87.68	**87.86**	85.58	85.64	86.64	86.50
	AA	88.63	**89.94**	89.16	89.30	89.21	88.85
	κ	84.02	**84.28**	81.48	81.57	82.99	82.51

TABLE III

OA (%), AA (%), AND κ (%) OF THE PROPOSED GRAPHGST VERSUS THE NUMBER OF TRANSFORMER ENCODERS n ON ALL DATASETS

Dataset	n	1	3	5	7	9	11
Indian Pines	OA	96.40	96.57	96.98	96.61	**97.06**	96.73
	AA	97.77	98.04	98.32	98.09	**98.39**	98.21
	κ	95.87	96.07	96.54	96.11	**96.64**	96.26
WHU-Hi-LongKou	OA	95.99	**96.15**	95.97	96.11	95.77	95.85
	AA	96.64	**96.88**	96.51	96.59	96.34	96.33
	κ	94.78	**95.00**	94.76	94.94	94.50	94.60
Houston2013	OA	**90.81**	90.80	90.21	89.28	88.92	88.60
	AA	91.87	**91.93**	91.50	90.39	90.55	89.84
	κ	90.00	**90.02**	89.37	88.37	87.98	87.63
Houston2018	OA	85.64	86.68	86.94	87.15	**88.00**	87.50
	AA	87.14	89.41	89.95	89.76	**90.58**	90.02
	κ	81.07	82.89	83.17	83.44	**84.20**	84.13

2) Impact of Threshold Value γ: The threshold value γ determines the criteria for two pixels to be considered as positive pairs in the GCCVL module. To investigate the impact of γ on the classification performance of GraphGST, we conduct experiments on four datasets and examine the metrics (OA, AA, and κ) as γ varies. Table II records the experimental results. We adopt a batch-wise strategy to train the model, and the pixels of each batch are randomly selected from the training/testing set. Consequently, the number of adjacent pixels in a batch is limited. To ensure a sufficient number of positive samples, we set γ within the range of $[0, 0.1, \ldots, 0.5]$. Table II shows that these metrics first increase and then decrease as γ increases on four datasets. This is because many pixels at greater distances are treated as positive sample pairs when the threshold is too low. However, these pixels at greater distances

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TABLE IV

CLASSIFICATION ACCURACIES (%) OBTAINED BY DIFFERENT APPROACHES ON THE INDIAN PINES DATASET. CAS ARE LISTED FOR COMPARISON. THE BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	NSCKL	FuNet	HiT	SpectralFormer	MAEST	morphFormer	SSFTT	GraphGST
CA	Corn Notill	50	1384	92.80	68.50	70.44	70.52	78.97	85.23	91.18	**95.81**
	Corn Mintill	50	784	96.81	79.59	85.96	81.89	92.73	85.31	98.72	**98.85**
	Corn	50	184	98.17	99.46	93.47	91.30	98.91	92.65	100.00	**100.00**
	Grass Pasture	50	447	86.08	95.08	93.95	95.53	96.20	88.65	96.19	**97.09**
	Grass Trees	50	697	99.57	95.70	89.09	85.51	89.10	96.73	**100.00**	98.57
	Hay Windrowed	50	439	100.00	99.54	94.08	99.32	98.41	94.84	**100.00**	99.54
	Soybean Notill	50	918	92.96	75.93	80.39	81.81	84.86	83.17	95.10	**97.93**
	Soybean Mintill	50	2418	**98.40**	68.90	68.85	75.48	73.44	81.01	94.44	94.50
	Soybean Clean	50	564	94.28	71.63	70.03	73.76	69.50	74.24	90.96	**95.04**
	Wheat	50	162	100.00	99.38	98.14	98.77	100.00	95.11	**100.00**	98.76
	Woods	50	1244	**99.75**	89.55	93.64	93.17	92.12	91.12	99.35	99.59
	Buildings Grass Tress Drives	50	330	93.56	91.52	82.12	78.48	91.51	85.73	**100.00**	98.48
	Stone Steel Towers	50	45	93.26	100.00	97.77	100.00	100.00	96.71	100.00	**100.00**
	Alfalfa	15	39	98.00	94.87	74.35	79.49	89.74	89.22	100.00	**100.00**
	Grass Pasture Mowed	15	11	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**
	Oats	15	5	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**
OA				96.28	79.89	80.21	81.76	84.15	93.77	96.11	**97.06**
AA				96.48	89.35	87.02	87.81	90.97	91.03	97.92	**98.39**
κ				95.76	77.16	77.54	79.19	82.00	90.71	95.55	**96.64**

TABLE V

CLASSIFICATION ACCURACIES (%) OBTAINED BY DIFFERENT APPROACHES ON THE WHU-HI-LONGKOU DATASET. CAS ARE LISTED FOR COMPARISON. THE BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	NSCKL	FuNet	HiT	SpectralFormer	MAEST	morphFormer	SSFTT	GraphGST
CA	Corn	25	34486	99.79	97.82	93.28	95.80	98.52	96.14	**98.33**	97.89
	Cotton	25	8349	96.45	45.71	77.25	65.61	63.06	93.28	**96.90**	95.87
	Sesame	25	3006	97.60	87.25	89.17	76.74	90.48	98.36	**99.53**	99.33
	Broad-Leaf Soybean	25	63187	86.67	81.06	79.75	81.77	90.36	85.78	89.63	**90.42**
	Narrow-Leaf Soybean	25	4126	95.77	88.87	88.36	80.92	70.62	91.03	85.79	**99.49**
	Rice	25	11829	99.64	95.02	87.70	95.22	97.50	99.53	99.57	**100.00**
	Water	25	67031	99.95	98.79	99.92	99.59	99.50	99.82	97.27	**99.97**
	Roads and Houses	25	7099	94.54	85.16	89.60	81.94	86.09	92.92	93.49	**95.07**
	Mixed Weed	25	5204	**95.53**	92.25	89.27	85.53	85.99	90.94	89.63	93.89
OA				95.24	89.75	89.90	90.12	93.38	93.91	95.24	**96.15**
AA				96.22	85.78	88.25	84.79	86.91	94.20	95.44	**96.88**
κ				93.83	86.80	87.03	87.25	91.38	92.12	94.46	**95.00**

TABLE VI

CLASSIFICATION ACCURACIES (%) OBTAINED BY DIFFERENT APPROACHES ON THE HOUSTON UNIVERSITY 2013 DATASET. CAS ARE LISTED FOR COMPARISON. THE BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	NSCKL	FuNet	HiT	SpectralFormer	MAEST	morphFormer	SSFTT	GraphGST
CA	Healthy-Grass	198	1053	93.35	83.86	**95.15**	81.86	82.90	93.63	85.37	85.66
	Stressed-Grass	190	1064	99.62	98.59	98.12	**100.00**	99.44	98.02	99.81	99.91
	Synthetic-Grass	192	505	95.64	83.37	**99.20**	95.25	94.65	99.20	95.05	95.64
	Trees	188	1056	94.03	**98.96**	96.30	96.12	96.02	95.64	90.24	91.29
	Soil	186	1056	100.00	99.72	100.00	99.53	99.24	98.95	100.00	**100.00**
	Water	182	143	100.00	96.50	100.00	94.41	93.01	100.00	100.00	**100.00**
	Residential	196	1072	84.70	89.55	86.56	83.12	**89.93**	86.66	75.00	76.96
	Commercial	191	1053	**91.45**	89.36	84.61	76.73	82.15	83.85	89.74	89.84
	Road	193	1059	77.71	83.29	70.72	79.32	77.15	75.92	76.86	**83.57**
	Highway	191	1036	78.66	79.25	61.96	78.86	87.55	66.50	90.92	**91.79**
	Railway	181	1054	74.19	79.89	83.01	88.71	85.58	82.06	87.95	**91.46**
	Parking-Lot1	192	1041	**93.17**	79.15	82.22	87.32	80.60	87.22	92.02	90.49
	Parking-Lot2	184	285	81.05	**87.72**	77.89	72.63	69.82	84.91	80.00	81.40
	Tennis Court	181	247	100.00	93.93	100.00	100.00	97.17	100.00	100.00	**100.00**
	Buildings-Grass-Tress	187	473	100.00	98.94	99.58	99.79	98.10	99.57	100.00	**100.00**
OA				89.61	88.62	87.27	88.01	88.55	88.27	89.62	**90.80**
AA				90.91	89.47	89.03	88.91	88.89	90.15	90.87	**91.93**
κ				88.72	87.64	86.1	86.99	87.57	87.27	88.73	**90.02**

are likely to belong to different classes. Simultaneously, the occurrence of too few positive sample pairs may result in the failure of the self-supervised signal when a high threshold is set. Nevertheless, the accuracies are not significantly affected by variations in γ. This indicates the robustness of our model. Finally, to ensure the best performance, we set $\gamma = 0.1$ for the Indian Pines and Houston University 2018 datasets and $\gamma = 0.2$ for the WHU-Hi-LongKou and Houston University 2013 datasets.

3) Impact of Number of Transformer Encoders n: Increasing the number of Transformer encoders n can boost the model's capacity, allowing it to represent more complex

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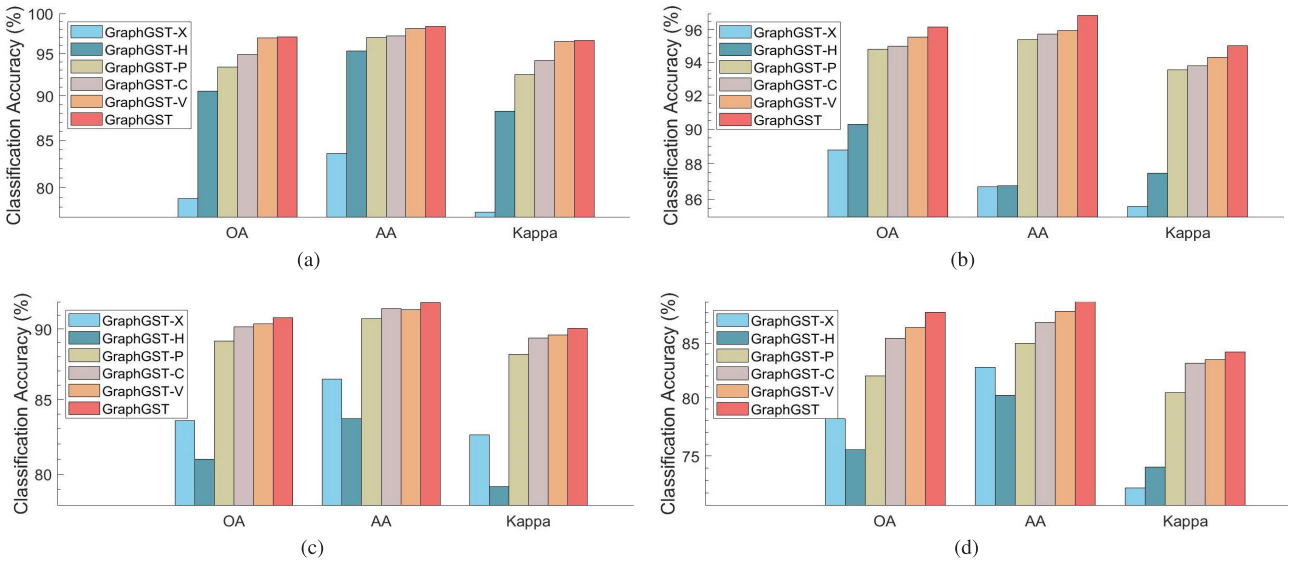


Fig. 6. Experimental results of GraphGST and its five variants on different datasets. (a) Indian Pines dataset. (b) WHU-Hi-LongKou. (c) Houston University 2013. (d) Houston University 2018.

TABLE VII

CLASSIFICATION ACCURACIES (%) OBTAINED BY DIFFERENT APPROACHES ON THE HOUSTON UNIVERSITY 2018 DATASET. CAS ARE LISTED FOR COMPARISON. THE BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	NSCKL	FuNet	HiT	SpectralFormer	MAEST	morphFormer	SSFTT	GraphGST
CA	Healthy Grass	100	9395	73.53	82.69	85.55	82.95	85.09	90.28	73.66	**92.62**
	Stressed Grass	500	31430	**96.76**	94.34	95.11	95.17	95.42	91.78	92.65	91.76
	Artificial Turf	100	544	99.21	99.15	99.05	99.00	99.81	99.10	99.63	**99.82**
	Evergreen Trees	500	12910	**99.60**	97.86	97.98	97.73	98.63	97.58	98.52	97.88
	Deciduous Trees	100	4828	89.23	90.03	88.44	87.46	89.93	88.56	90.26	**92.71**
	Bare Earth	100	4383	98.61	98.31	98.40	98.74	98.79	98.68	99.88	**100.00**
	Water	50	188	94.68	97.87	98.93	97.87	97.87	100.00	100.00	**100.00**
	Residential Buildings	500	39005	**96.06**	88.45	90.87	91.15	92.29	84.97	86.72	91.15
	Non-Residential Buildings	500	223496	86.68	85.25	83.97	84.65	85.34	89.48	91.51	**93.48**
	Roads	500	44444	72.46	68.41	70.87	**72.61**	69.94	65.53	68.55	67.04
	Sidewalks	500	31633	**76.68**	66.02	73.74	72.97	69.25	73.77	62.85	65.04
	Crosswalks	100	1417	48.20	40.93	60.76	38.88	52.78	61.86	49.32	**62.87**
	Major Thoroughfares	500	46283	**87.77**	83.36	67.64	76.01	80.00	72.85	82.85	79.93
	Highway	100	9782	83.18	81.01	86.55	84.63	81.71	79.33	91.36	**91.93**
	Railways	100	6827	**99.17**	98.68	96.86	99.06	98.62	98.44	92.36	96.74
	Paved Parking Lots	500	10937	**98.99**	98.18	97.64	96.35	96.85	96.80	96.12	95.51
	Unpaved Parking Lots	50	89	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**
	Cars	100	6505	84.20	81.29	87.79	87.34	76.52	84.81	90.06	**92.09**
	Trains	100	5129	96.33	95.14	97.09	95.06	**97.44**	95.80	97.05	96.72
	Stadium Seats	100	6690	98.86	99.20	99.86	99.56	98.35	98.66	99.94	**100.00**
OA				86.98	84.13	83.33	84.36	84.61	85.03	86.58	**88.00**
AA				89.05	87.35	88.91	87.91	88.24	88.41	88.17	**90.58**
κ				83.41	79.81	78.91	80.13	80.42	81.03	82.70	**84.20**

functions and capture finer-grained patterns (thus enhancing the model's performance). However, too many transformer encoders can result in a high computational burden and can also lead to problems such as gradient disappearance and overfitting. To understand the influence of n on the experimental results of our GraphGST, we conduct experiments on four datasets and evaluate three metrics (OA, AA, and κ) across different values of n. Table III records the experimental results. Herein, we set n in the ranges of $[1, 3, \ldots, 11]$. As observed, the variations in n do not markedly impact these metrics, highlighting the robustness of our model. However, the impact of increasing n on the model's performance varies across different datasets. Specifically, our GraphGST achieves the best performance when n is set to 9 (Indian Pines and Houston 2018 datasets). In WHU-Hi-LongKou and Houston University 2013 datasets, our GraphGST performs best when n equals 3. Finally, we set $n = 9$ for the Indian Pines and Houston 2018 datasets and $n = 3$ for the WHU-Hi-LongKou and Houston University 2013 datasets.

D. Ablation Study

In this section, we compare the proposed GraphGST with its five variants to assess the necessity and effectiveness of each component involved in the proposed GraphGST. GraphGST incorporates multiple components, including graph structure embedding, classic Transformer encoders, the positional encoder, and the GCCVL module. Accordingly, we design five variants of GraphGST. By comparing GraphGST with its variants, we aim to understand the contributions of each

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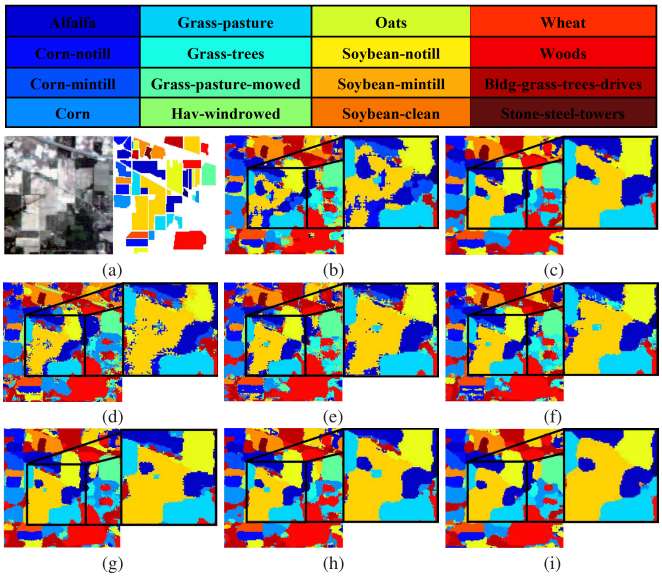


Fig. 7. Classification maps obtained by different methods on the Indian Pines dataset. (a) RGB and GT. (b) FuNet. (c) NSCKL. (d) HiT. (e) SpectralFormer. (f) MAEST. (g) morphFormer. (h) SSFTT. (i) GraphGST.

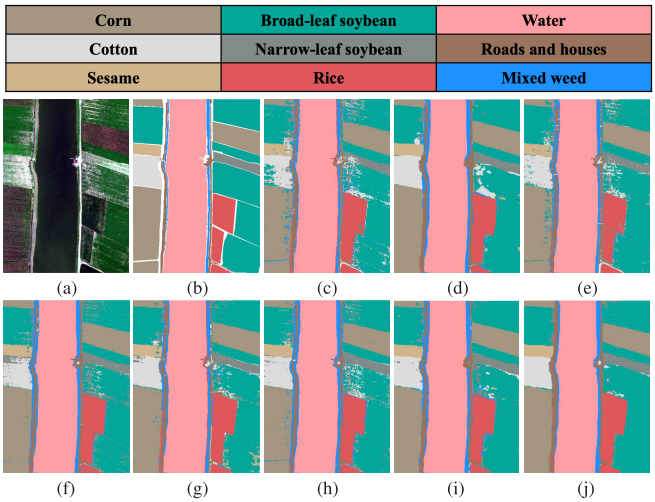


Fig. 8. Classification maps obtained by different methods on the WHU-Hi–LongKou dataset. (a) RGB. (b) GT. (c) FuNet. (d) NSCKL. (e) HiT. (f) SpectralFormer. (g) MAEST. (h) morphFormer. (i) SSFTT. (j) GraphGST.

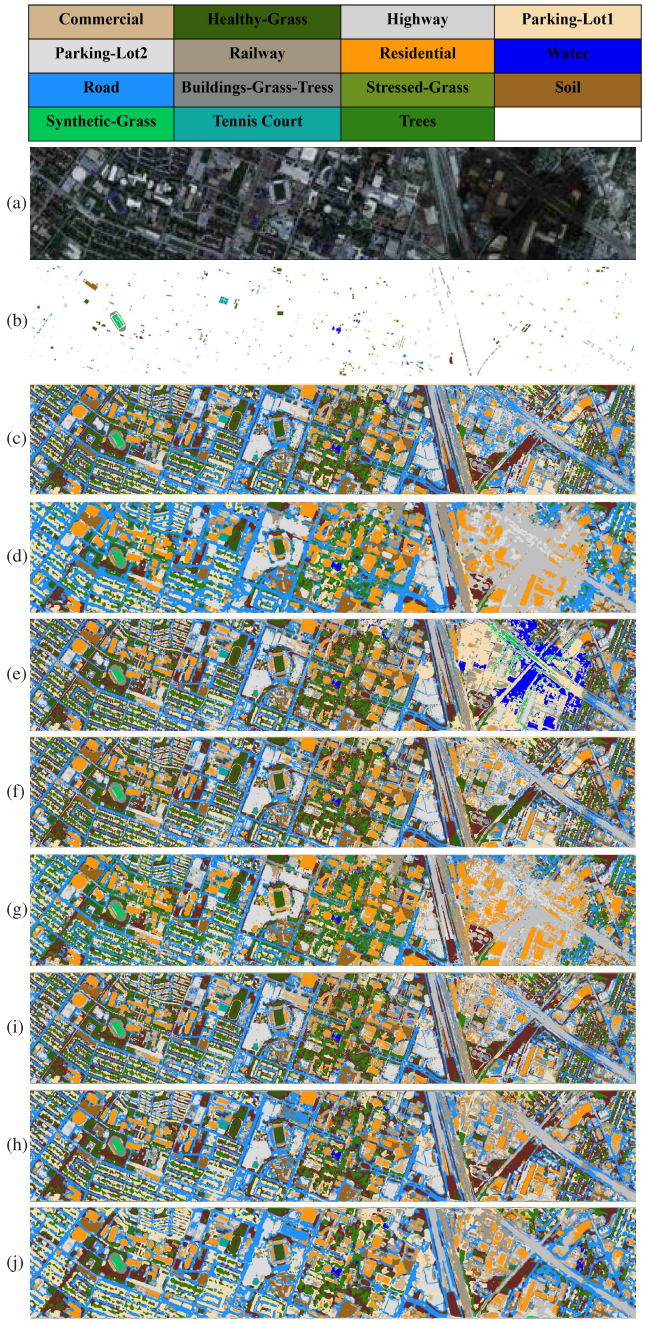


Fig. 9. Classification maps obtained by different methods on the Houston University 2013 dataset. (a) RGB. (b) GT. (c) FuNet. (d) NSCKL. (e) HiT. (f) SpectralFormer. (g) MAEST. (i) morphFormer. (h) SSFTT. (j) GraphGST.

component to the overall performance. The five variants are represented as follows.

1) *GraphGST-X:* A variant of GraphGST that does not learn GSEs of pixels. Thus, the input of the Encoder 1 is given by the normalized spectral features of pixels $\mathbf{X}$.
2) *GraphGST-H:* A variant of GraphGST that does not consider all Transformer encoders. From a practical perspective, the high-level feature representations of pixels $\overrightarrow{\mathbf{H}} = \mathbf{H}^{(0)}$.
3) *GraphGST-P:* A variant of GraphGST that does not employ classic Transformer encoders to learn pixel representations. From a practical perspective, the GSEs of pixels $\mathbf{H}^{(0)}$ and APSs of pixels $\widetilde{\mathbf{H}}$ are taken as the input of the positional encoder, and the high-level feature representations of pixels $\overrightarrow{\mathbf{H}} = \|(\mathbf{H}^{(0)}, \widetilde{\mathbf{H}})$.
4) *GraphGST-C:* A variant of GraphGST that does not use the positional encoder to capture locational correlations between pixels. From a practical perspective, the high-level feature representations of pixels $\overrightarrow{\mathbf{H}} = \|(\mathbf{H}^{(0)}, \mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(n)})$.
5) *GraphGST-V:* A variant of GraphGST that does not consider the GCCVL module to enhance the representation learning. From a practical perspective, the total loss $\ell = \ell_{\mathrm{CE}}$.

Fig. 6 presents the results of GraphGST and its five variants in terms of OA, AA, and κ on the four considered datasets. Obviously, the classification accuracies for GraphGST are

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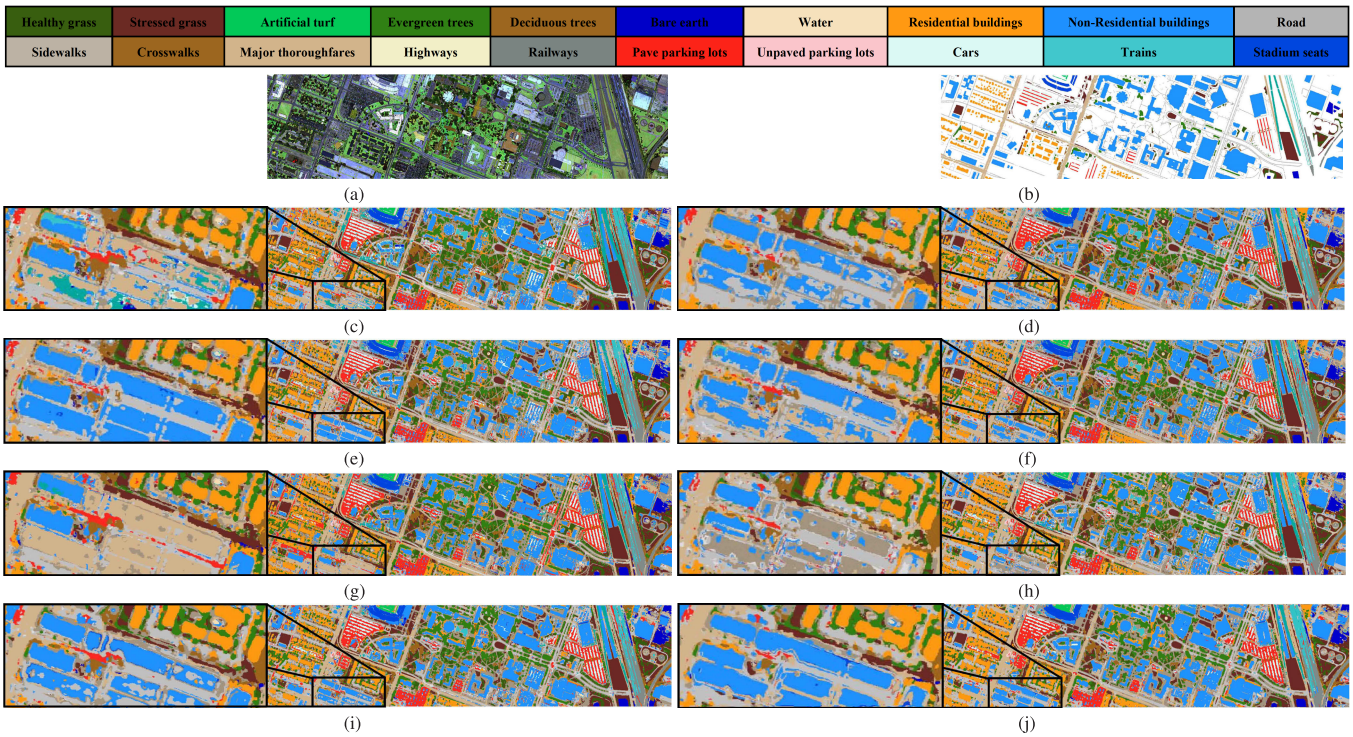


Fig. 10. Classification maps obtained by different methods on the Houston University 2018 dataset. (a) RGB. (b) GT. (c) FuNet. (d) NSCKL. (e) HiT. (f) SpectralFormer. (g) MAEST. (h) morphFormer. (i) SSFTT. (j) GraphGST.

higher than those five variants, indicating the effectiveness of graph structure embedding, classic Transformer encoders, the positional encoder, and the GCCVL module. By comparing GraphGST-X and GraphGST-P, we observe that GraphGST-P achieves better prediction performances. This finding implies that both local and global correlations are significant for the HSI classification task, while local correlation is more crucial. Comparing the results of GraphGST-H and GraphGST-P, we can see that the metrics provided by GraphGST-P are higher than those by GraphGST-H. This finding confirms the effectiveness of the positional encoder, and it can be viewed as a complement to graph structure embedding in capturing local correlations. In our GraphGST, pixel features under the local and global views are derived from GSEs, making these three feature vectors possibly share some common subspace. GraphGST-X, GraphGST-P, and GraphGST-C can be interpreted as configurations omitting GSEs, global view, and local view features, respectively. Upon comparing the results, it becomes evident that GraphGST-X, GraphGST-P, and GraphGST-C display suboptimal performance, with GraphGST-X being particularly noteworthy. Therefore, we can confidently assert that the combination of graph structure embedding and the global and local view features does not cause inefficiencies in accuracy. This is attributed to the fact that these features focus on distinct pixel relationships, making them inherently unique and complementary.

E. Comparative Experiments

To further assess the performance of GraphGST, four metrics (CA, OA, AA, and κ) are presented in Tables IV to VII, offering an evaluation of the classification performance of the proposed GraphGST and seven state-of-the-art (SOTA) baselines (NSCKL, FuNet, HiT, SpectralFormer, MAEST, morphFormer, and SSFTT). The evaluation is conducted on four datasets: Indian Pines, WHU-Hi-LongKou, Houston University 2013, and Houston University 2018.

Among these selected methods, NSCKL and FuNet are graph models, while the rest of the baselines are Transformer-based DL approaches. In particular, HiT, SpectralFormer, MAEST, and morphFormer are ViT-based approaches. SSFTT and the proposed GraphGST are the classic Transformer-based approaches. It can be observed that the NSCKL achieves good classification results using all four datasets, which is attributed to the fact that NSCKL hierarchically develops a kernel-based iterative filter and fast spectral clustering, enabling it to capture local-global dependencies of the HSI. Meanwhile, the four ViT-based approaches (HiT, SpectralFormer, MAEST, and morphFormer) achieve similar classification performance on three datasets (except Indian Pines). Compared to the four ViT-based approaches, the two classic Transformer-based approaches (SSFTT and the proposed GraphGST) perform better, especially GraphGST. This is mainly due to the fact that ViT-based approaches take patches as tokens and adopt a patch-to-patch strategy to implement training, which prevents them from distinguishing heterogeneous pixels within the same patch. Although the classic Transformer lacks the ability to capture local features, it can be integrated with local feature extractors to achieve good classification results. Specifically, SSFTT employs a 3-D convolution layer and a 2-D convolution layer to extract local features, and our GraphGST constructs a pixel graph and uses MPNN to learn graph structural features for pixels.

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Moreover, we present the APE method to make the transformer aware of the locational relationship between pixels. In the Indian Pines dataset, our GraphGST exhibits 15.6%, 10.58%, and 17.45% performance improvements over SpectralFormer in terms of OA, AA, and κ, respectively. Additionally, our GraphGST achieves 1.28%, 1.77%, and 1.58% performance improvements over the SSFTT in terms of OA, AA, and κ, respectively, in the Houston University 2018 dataset. Overall, our GraphGST achieves the best performance in most cases.

Furthermore, we visualize the classification maps obtained by different methods for a qualitative evaluation. Figs. 7–10 display the classification maps obtained for the Indian Pines, WHU-Hi-LongKou, Houston University 2013, and Houston University 2018 datasets. In this article, all the baselines and our GraphGST can extract the local features of HSIs. Thus, the salt and pepper noise in classification maps can be reduced to some extent. Nevertheless, FuNet and four ViT-based approaches still perform unsatisfactorily. This indirectly indicates that these approaches fail to classify objects accurately. In general, taking patches as tokens prevents the model from distinguishing heterogeneous pixels within the same patch, leading to classification failure. Moreover, taking patches as tokens also increases the computational burden. For this reason, the patch size is often set relatively small. Consequently, these ViT-based approaches cannot aggregate sufficient local features. In contrast, it can be observed that our GraphGST produces the smoothest classification maps and accurately retains edge information.

V. Conclusion

NE and PE are two indispensable components in a Transformer, where the former excavates hidden data correlations while the latter stores locational relationships between nodes. However, in achieving HSI classification, most existing Transformer-based methods randomly generate PE, which makes it difficult to achieve strong representations. This article proposes a GraphGST classifier that solves the above-mentioned PE problem. Specifically, our GraphGST designs an APE to obtain APSs for pixels, enabling the Transformer to recognize the locational information of pixels in an HSI. Meanwhile, GSE is integrated into an MHSA mechanism for capturing the structural interaction between pixels. Moreover, a generative mechanism with self-supervised learning is developed to achieve cross-view CL, aiming to enhance the representation learning of pixels. By the aforementioned procedures, the proposed GraphGST model obtains local-to-global correlations for improving classification tasks. The classification performance of the GraphGST model is evaluated on four real datasets, obtaining SOTA results. Our future work will improve the current GCCVL scheme to enhance the inferential capabilities of the network further.

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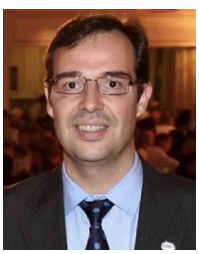
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